

Mac Development Setup Guide - Induction Racket Application

This document outlines the complete setup process for running the Induction Racket application on macOS, including database, backend, and frontend configuration.

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Prerequisites

Before starting, ensure you have the following installed:

- **Homebrew** (macOS package manager)
- **Node.js** and **npm** (for frontend)
- **Python 3.x** (for backend)
- **Git** (for version control)

To install Homebrew if not already installed:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

MySQL Database Setup

1. Install MySQL using Homebrew

```
brew install mysql
```

2. Start MySQL Service

```
brew services start mysql
```

3. Verify MySQL Installation

```
mysql --version
```

4. Create Database (if needed)

```
mysql -u root -p
```

Then in the MySQL prompt:

```
CREATE DATABASE proof_buddy;  
EXIT;
```

Backend Setup

1. Navigate to Backend Directory

```
cd /path/to/your/project/backend
```

2. Create Python Virtual Environment

```
python3 -m venv venv
```

3. Activate Virtual Environment

```
source venv/bin/activate
```

Note: You should see (venv) prefix in your terminal prompt

4. Install Python Dependencies

```
pip install -r requirements.txt
```

6. Run Database Migrations (if applicable)

```
python manage.py migrate
```

7. Start Backend Server

```
python manage.py runserver
```

Frontend Setup

1. Navigate to Frontend Directory

```
cd /path/to/your/project/frontend
```

2. Install Node Dependencies

```
npm install
```

3. Install Testing Dependencies (if not included)

```
npm install --save-dev @testing-library/react @testing-library/jest-dom @testing-library/user-event
```

4. Configure API Endpoint

Ensure your frontend is configured to connect to the backend. Check configuration files like:

- `src/config.js`
- `.env` or `.env.local`

Example `.env.local`:

```
REACT_APP_API_URL=http://localhost:5000
```

5. Start Frontend Development Server

```
npm start
```

The application should open automatically in your browser at <http://localhost:3000>

Running Tests

1. Run InductionRacket Component Tests

```
npm test src/pages/InductionRacket.test.js
```

2. Run All Tests

```
npm test
```

3. Run Tests in Watch Mode

```
npm test -- --watch
```

4. Run Tests with Coverage

```
npm test -- --coverage
```

Verification

Verify MySQL is Running

```
brew services list | grep mysql
```

Should show **started** status

Verify Frontend is Running

Open browser to <http://localhost:3000> and check:

- Page loads without errors
- "Induction: Racket" header is visible
- All form fields are present (Name, # Tag, IVar, AVal, LVar)
- Console has no errors

Verify Tests Pass

All tests in `InductionRacket.test.js` should pass:

- Component rendering tests
 - Form input handling tests
 - Validation logic tests
 - Button interaction tests
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Troubleshooting

MySQL Connection Issues

Problem: Cannot connect to MySQL

```
# Check if MySQL is running
brew services list

# Restart MySQL
brew services restart mysql

# Check MySQL logs
tail -f /usr/local/var/mysql/*.err
```

Virtual Environment Issues

Problem: Cannot activate virtual environment

```
# Recreate virtual environment
rm -rf venv
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

Port Already in Use

Problem: Port 3000 or 5000 already in use

```
# Find process using port
lsof -i :3000
lsof -i :5000

# Kill process
kill -9 <PID>
```

Node Module Issues

Problem: Module not found errors

```
# Clear npm cache and reinstall  
rm -rf node_modules package-lock.json  
npm cache clean --force  
npm install
```

Test Failures

Problem: Tests failing unexpectedly

```
# Clear Jest cache  
npm test -- --clearCache  
  
# Run tests in verbose mode  
npm test -- --verbose
```

Common Commands Reference

Backend

```
# Activate virtual environment  
source venv/bin/activate  
  
# Deactivate virtual environment  
deactivate  
  
# Run backend server  
python app.py
```

Frontend

```
# Start development server  
npm start  
  
# Run tests  
npm test  
  
# Build for production  
npm run build
```

MySQL

```
# Start MySQL
brew services start mysql

# Stop MySQL
brew services stop mysql

# Access MySQL CLI
mysql -u root -p
```

Notes

- Always activate the virtual environment before working on backend code
 - Keep dependencies up to date in `requirements.txt` and `package.json`
 - Commit `.env.example` files but never commit actual `.env` files with credentials
 - Run tests before committing changes
 - Use `brew services` to manage MySQL instead of manual starts/stops
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